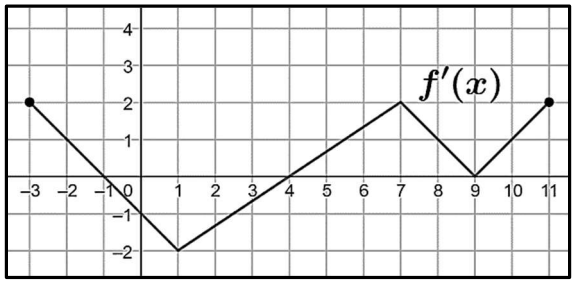
Worksheet 5-4 A Relative Extrema #1
AP Calculus

Name SOLUTIONS

Directions: Use the following information to answer the given questions. Show all work.



1. The graph of f', the derivative of f, is shown above for $[-3, 11]$ and consists of four line segments. Use the graph of f' to answer the following.

A) Find all critical values of $f(x)$ in the open interval $(-3, 11)$.

$x=-1$, $x=4$, $x=9$ $\quad$ $f'(x)=0$ at these values

B) Does $f(x)$ have a relative minimum, relative maximum, or neither at $x = 7$? Give a reason for your answer.

$f(x)$ has no relative extrema at $x = 7$ because $f'(7) \neq 0$ and $f'(x)$ does not change signs at $x = 7$

C) Does $f(x)$ have a relative minimum, relative maximum, or neither at $x = -1$? Give a reason for your answer.

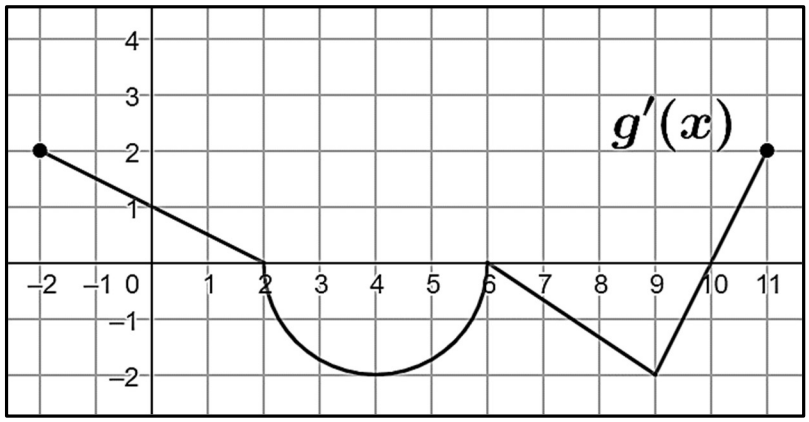
$f(-1)$ is a relative maximum because $f'(-1) = 0$ and $f'(x)$ changes signs from + to – at $x = -1$

D) Does $f(x)$ have a relative minimum, relative maximum, or neither at $x = 9$? Give a reason for your answer.

$f(x)$ has no relative extrema at $x = 9$ because $f'(9) = 0$ and $f'(x)$ does not change signs at $x = 7$

E) On what interval(s) in $[-3, 11]$ is $f(x)$ increasing? Give a reason for your answer.

$f(x)$ is increasing on $[-3,-1]$ and $[4,11]$ because $f'(x) \geq 0$ on these intervals



2. The graph of g', the derivative of g, is shown above for $[-2, 11]$ and consists of three line segments and a semicircle. Use the graph of g' to answer the following.

A) Find all critical values of $g(x)$ in the open interval $(-2, 11)$.

$X=2$, $X=6$, $X=10$ $\quad$ $f'(x)=0$ at these points

B) At which value(s) of x does $g(x)$ have a local minimum? Explain your reasoning.

$g(10)$ is a local minimum because $g'(10) = 0$ and $g'(x)$ changes from $-$ to $+$ at $x = 10$

C) On what interval(s) in $[-2, 11]$ is $g(x)$ increasing? Give a reason for your answer

$g(x)$ is increasing on $[-2, 2]$ and $[10, 11]$ because $g'(x) \geq 0$ on these intervals

D) Does $g(x)$ have a local minimum, local maximum, or neither at $x = 4$? Give a reason for your answer.

$g(x)$ has no relative extrema at $x = 4$ because $g'(4) \neq 0$ and $g'(x)$ does not change signs at $x = 4$

E) Let $h(x) = xg(x)$. If $h(x)$ has a relative maximum at $x = 4$, find $h(4)$. Show the work that leads to your answer.

$h(x)$ has a rel. max at $x=4 \rightarrow h'(4)=0$

$h'(x) = 1 \cdot g(x) + x \cdot g'(x)$

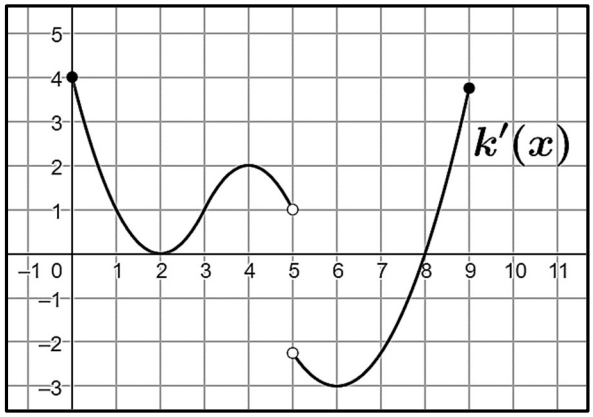
$g(4) + 4g'(4) = 0$

$g(4) + 4(-2) = 0$

$g(4) = 8$

$h(4) = 4 \cdot g(4)$

$= 4(8) = 32$



3. The function k is continuous on the interval $[0, 9]$. The graph of $k'(x)$ is given above and has horizontal tangents at $x = 2, x = 4,$ and $x = 6$.

A) On which interval(s) is $k(x)$ increasing? Give a reason for your answer.

$k(x)$ is increasing on [0, 5) and [8, 9] because $k'(x) \geq 0$ on these intervals

B) At which value(s) of x does $k(x)$ have a relative minimum? Explain your reasoning.

$k(8)$ is a relative minimum because $k'(8) = 0$ and $k'(x)$ changes from – to + at $x = 8$

C) At which value(s) of x does $k(x)$ have a local maximum? Explain your reasoning.

$k(5)$ is a relative maximum because $k'(5)$ is undefined and $k'(x)$ changes from + to – at $x = 5$

D) Let $m(x) = k(x) - \sin(x - 2)$. Find $m'(2)$. Does $m(x)$ have a local minimum, local maximum, or neither at $x = 4$? Give a reason for your answer.

$m'(x) = K'(x) - \cos(x-2)$

$m'(2) = K'(2) - \cos(2-2) = 0 - \cos 0 = 0 - 1 = -1$

$m'(4) = K'(4) - \cos(4-2) = 2 - \cos 2 \neq 0$

$m(x)$ has no relative extrema at $x=4$ because $m'(4) \neq 0$

Directions: Given the following information, find and classify the x-values of any relative extrema of the given function. Justify your answers.

4. $f(x) = -x^3 + 6x^2 - 9x + 7$

$f'(x) = -3x^2 + 12x - 9 = 0$

$-3(x^2 - 4x + 3) = 0$

$-3(x-1)(x-3) = 0$

$x = 1 \quad x = 3$

f': $-$ | $+$ | $-$

f: $\searrow$ 1 $\nearrow$ 3 $\searrow$

$f(1)$ is a relative minimum because $f'(1) = 0$ and $f'(x)$ changes signs from – to + at $x = 1$.

$f(3)$ is a relative maximum because $f'(3) = 0$ and $f'(x)$ changes signs from + to – at $x = 3$.

5. $f'(x) = 3x(x - 4)(x + 1)$

$f'(x) = 0 \rightarrow 3x(x-4)(x+1) = 0$

$x = 0 \quad x = 4 \quad x = -1$

f': $-$ | $+$ | $-$ | $+$

f: $\searrow$ -1 $\nearrow$ 0 $\searrow$ 4 $\nearrow$

$f(-1)$ and $f(4)$ are relative minimums because $f'(-1) = 0, f'(4) = 0$, and $f'(x)$ changes signs from – to + at $x = -1$ and $x = 4$.

$f(0)$ is a relative maximum because $f'(0) = 0$ and $f'(x)$ changes signs from + to – at $x = 0$.

6. $f(x) = x^2 e^x$

$f'(x) = 2xe^x + x^2e^x = 0$

$xe^x(2+x) = 0$

$x = 0 \quad e^x = 0$ no sol. $\quad 2 + x = 0$, $x = -2$

$f(0)$ is a relative minimum because $f'(0) = 0$ and $f'(x)$ changes signs from – to + at $x = 0$.

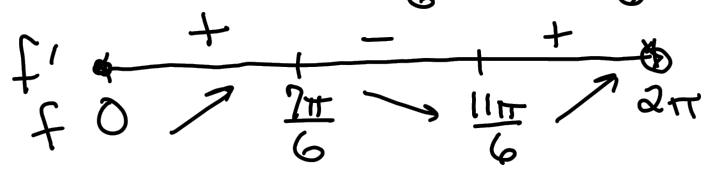
$f(-2)$ is a relative maximum because $f'(-2) = 0$ and $f'(x)$ changes signs from + to – at $x = -2$.

7. $f'(x) = 2\sin(x) + 1$ on $[0, 2\pi)$

$f'(x) = 2\sin x + 1 = 0$

$\sin x = -\frac{1}{2}$

$x = \frac{7\pi}{6} \quad x = \frac{11\pi}{6}$



$f\left(\frac{11\pi}{6}\right)$ is a relative minimum because $f'\left(\frac{11}{6}\right) = 0$ and $f'(x)$ changes signs from – to + at $x = \frac{11}{6}$.

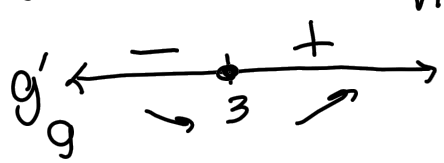
$f\left(\frac{7\pi}{6}\right)$ is a relative maximum because $f'\left(\frac{7\pi}{6}\right) = 0$ and $f'(x)$ changes signs from + to – at $x = \frac{7\pi}{6}$.

8. $g'(x) = \frac{x-3}{e^x}$

$\{e^x > 0$ for all x

$g'(x) = 0$ when $x - 3 = 0$, $x = 3$

$g'(x)$ is undef when $e^x = 0$ no sol



$g(3)$ is a relative minimum because $g'(3) = 0$ and $g'(x)$ changes signs from – to + at $x = 3$.

9. $g(x) = \frac{x-3}{x^2}$

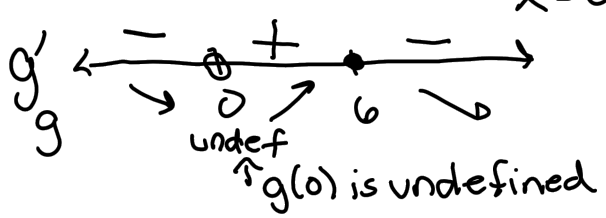
$g'(x) = \frac{x^2(1) - (x-3)(2x)}{x^4}$

$= \frac{x^2 - 2x^2 + 6x}{x^4}$

$= \frac{-x^2 + 6x}{x^4} = \frac{-x+6}{x^3}$

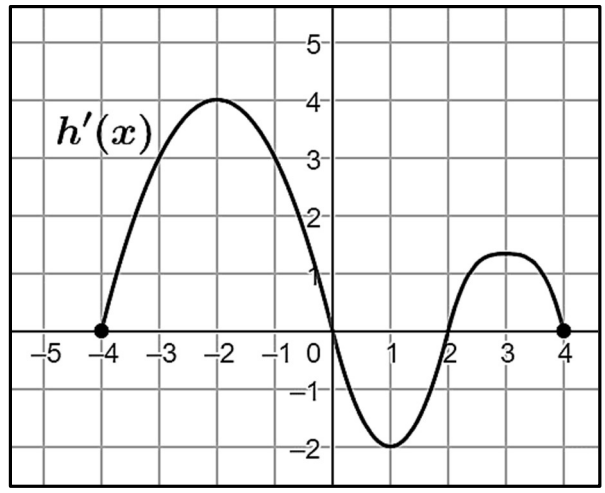
$g'(x) = 0$ when $-x + 6 = 0$, $x = 6$

$g'(x)$ undef when $x^3 = 0$, $x = 0$



$g(6)$ is a relative maximum because $g'(6) = 0$ and $g'(x)$ changes signs from + to – at $x = 6$.

10. The function h is twice differentiable on the interval $[-4, 4]$. The graph of $h'(x)$ is given at the right and has horizontal tangents at $x = -2, x = 1$, and $x = 3$. It is known that $h(-1) = 5$.



A) On which interval(s) is $h(x)$ decreasing? Give a reason for your answer.

$h(x)$ is decreasing on $[0, 2]$ because $h'(x) \leq 0$ on this interval

B) Find all critical values for the function $h(x)$. Classify each critical value as a local minimum, local maximum, or neither. Explain your reasoning.

Critical values are at $x = -4, x = 0, x = 2$, and $x = 4$ because $h'(x) = 0$ at these x-values

No relative extrema occur at $x = -4$ and $x = 4$ because they are the endpoints of the interval

$h(0)$ is a relative maximum because $h'(0) = 0$ and $h'(x)$ changes signs from + to – at $x = 0$.

C) Let $k(x) = h(x) - 3x$. At which x value(s) does $k(x)$ have a relative maximum? Explain your reasoning.

$k'(x) = h'(x) - 3$ $\quad k'(x) = 0$ when $h'(x) - 3 = 0 \rightarrow h'(x) = 3$

$x = -3$ & $x = -1$

k': $-$ on $(-4, -3)$, $+$ on $(-3, -1)$, $-$ on $(-1, 4)$

$-4 \searrow -3 \nearrow -1 \searrow 4$

$k(-1)$ is a relative maximum because $k'(-1) = 0$ and $k'(x)$ changes signs from + to – at $x = 0$.

D) Find the value of $\lim\limits_{x \to -1} \dfrac{h'(x^2) - 2x}{h(x) - 5}$ or state that it does not exist. Justify your answer.

$\lim\limits_{x \to -1} (h'(x^2) - 2x) = h'(1) + 2 = -2 + 2 = 0$ $\qquad h''(1) = 0$

$\lim\limits_{x \to -1} (h(x) - 5) = h(-1) - 5 = 5 - 5 = 0$

By L'Hospital's Rule:

$$\lim_{x \to -1} \frac{h'(x^2) - 2x}{h(x) - 5} = \lim_{x \to -1} \frac{h''(x^2) \cdot 2x - 2}{h'(x)} = \frac{h''(1)(-2) - 2}{h'(-1)}$$

$$= \frac{0 - 2}{3} = \frac{-2}{3}$$

x	$-4 \le x < -1$	-1	$-1 < x < 3$	3	$3 < x < 8$	8	$x > 8$
$f(x)$	negative	-8	negative	0	positive	2	positive
$f'(x)$	negative	0	positive	0	positive	undefined	negative

11. The function f is continuous on the interval $[-4, \infty)$. The table above gives selected information about f and its derivative f'. It is known that $f(x) < 3$ for all values of x in its domain.

A) On what interval(s) is the function $f(x)$ increasing? Give a reason for your answer.

$f(x)$ is increasing on $[-1, 8)$ because $f'(x) \ge 0$ on this interval

B) Find any critical values for the function $f(x)$. For each critical value, determine if it is the location of a relative minimum, a relative maximum, or neither for $f(x)$. Justify your answers.

Critical values are at $x = -1$ and $x = 3$ because $f'(x) = 0$ at these x-values and $x = 8$ because $f'(8)$ is undefined

No relative extrema occur at $x = 3$ because $f'(3) = 0$ but $f'(x)$ does not change signs at $x = 3$

$f(8)$ is a relative maximum because $f'(8)$ is undefined and $f'(x)$ changes signs from + to – at $x = 8$.

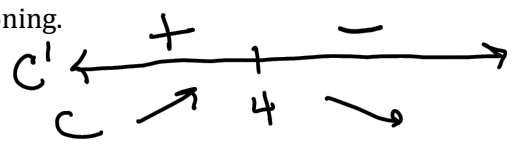
$f(-1)$ is a relative minimum because $f'(-1) = 0$ and $f'(x)$ changes signs from – to + at $x = 2$.

C) Let $C'(x) = (4 - x)f'(x)$ be the derivative of the function $C(x)$. Does C have a local minimum, local maximum, or neither at $x = 4$. Explain your reasoning.

$C'(4) = (4-4)f'(4) = 0$

$C'(3.5) = \underbrace{(4-3.5)}_{+}\underbrace{f'(3.5)}_{+} > 0$

$C'(4.5) = \underbrace{(4-4.5)}_{-}\underbrace{f'(4.5)}_{+} < 0$



$f'(x) > 0$ when $3 < x < 8$

$C(4)$ is a relative maximum because $C'(4) = 0$ and $C'(x)$ changes signs from + to – at $x = 4$.

x	$-4 \le x < -1$	-1	$-1 < x < 3$	3	$3 < x < 8$	8	$x > 8$
$f(x)$	negative	-8	negative	0	positive	2	positive
$f'(x)$	negative	0	positive	0	positive	undefined	negative

E) Let $n(x) = f(x^2)$. Does $n(x)$ have a local minimum, local maximum, or neither at $x = -1$. Justify your answer.

$n'(x) = f'(x^2) \cdot 2x$

$n'(-1) = \underbrace{f(1)}_{-} \cdot \underbrace{(-2)}_{-} > 0$

$n(x)$ has no relative extrema at $x = -1$ because $n'(-1) \neq 0$ and has no sign change at $x = -1$